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Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

AI Mini Project – I

AI-Based Plagiarism Detection and AI Content Analysis System

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
PDF	Portable Document Format

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ABSTRACT

In the digital era, the rapid growth of online content, academic resources, and AI-assisted writing tools has significantly increased concerns regarding plagiarism and the misuse of artificial intelligence in content creation. Educational institutions, research organizations, and industries are facing challenges in maintaining academic integrity and ensuring the originality of submitted documents. Traditional plagiarism detection systems primarily rely on exact text matching techniques, which compare documents against existing databases to identify copied content. However, these systems often fail to detect paraphrased text, reworded sentences, or content generated using advanced artificial intelligence tools. As a result, there is a growing need for more intelligent and reliable plagiarism detection mechanisms that can analyze the meaning and context of text rather than just matching words.

This project proposes an advanced AI-based plagiarism detection system that integrates semantic similarity analysis and AI-generated text detection to provide a more comprehensive evaluation of document originality. The system utilizes **Sentence Transformers**, a deep learning technique based on natural language processing (NLP), to measure the semantic similarity between documents. Unlike traditional methods, semantic similarity analysis focuses on understanding the meaning of sentences and identifying similarities even when the wording is different. This helps in detecting paraphrased content and hidden plagiarism more effectively.

In addition to semantic analysis, the system incorporates a **RoBERTa-based machine learning model** to detect whether a document has been generated by artificial intelligence tools. The RoBERTa model is trained on large datasets to recognize patterns commonly found in AI-generated text, such as repetitive structures, predictable language patterns, and unnatural phrasing. By combining both semantic similarity detection and AI text classification, the system provides a more accurate and reliable assessment of document authenticity.

The proposed system processes input documents through multiple stages, including text preprocessing, tokenization, feature extraction, and model evaluation. After analysis, the results are presented in a structured and user-friendly report that includes key metrics such as the similarity index, probability of AI-generated content, and an overall risk level indicating the likelihood of plagiarism or AI misuse. This automated reporting system helps educators, researchers, and organizations quickly evaluate document integrity and make informed decisions.

Overall, the proposed approach enhances the efficiency, accuracy, and reliability of plagiarism detection by leveraging advanced artificial intelligence and natural language processing techniques. It addresses the limitations of traditional plagiarism detection methods and provides a modern solution capable of identifying both paraphrased and AI-generated content, thereby supporting academic honesty and maintaining the quality of digital information.

Keyword-BMI (Body Mass Index), SVM (Super Vector Machines), KNN (K-Nearest Neighbours) Feature Selection, Logistic Regression.

CHAPTER 1
INTRODUCTION

In today's digital world, the creation and sharing of content have increased exponentially due to the widespread use of the internet and advanced technologies. Students, researchers, and professionals rely heavily on online resources for information, which has led to a significant rise in plagiarism. Plagiarism refers to the act of copying someone else's work, ideas, or expressions without proper acknowledgment. It is considered unethical and can have serious academic and professional consequences.

Traditional plagiarism detection systems primarily rely on exact text matching and database comparisons. While these systems are effective in identifying directly copied content, they often fail to detect paraphrased or reworded text. With the emergence of advanced Natural Language Processing (NLP) techniques, it has become possible to analyze the meaning and context of text rather than just matching words.

Another major challenge in recent years is the rise of AI-generated content. Modern language models are capable of producing highly coherent and human-like text, making it difficult to distinguish between human-written and machine-generated content. This has created new concerns in academic integrity, as AI tools can be misused to generate assignments, reports, and research papers.

To address these challenges, this project proposes an AI-based plagiarism detection system that combines semantic similarity analysis with AI content detection. The system uses transformer-based models to understand the context and meaning of sentences, allowing it to identify similarities even when the wording is different. Additionally, it incorporates an AI detection model to estimate the probability that a given text is generated by an AI system.

By integrating these two approaches, the proposed system provides a more comprehensive and accurate analysis of document originality. It not only detects similar content within a document but also evaluates the likelihood of AI-generated text, offering a detailed and informative report to the user.

CHAPTER 2
LITERATURE REVIEW

Plagiarism detection has been an active area of research for many years due to the increasing need to maintain academic integrity and originality in digital content. Traditional plagiarism detection methods mainly rely on string matching techniques such as fingerprinting, substring matching, and keyword comparison. These techniques compare a submitted document with a database of existing documents to identify copied text. Popular tools such as Turnitin and Copyscape use large online databases to detect similarities between documents and identify direct copying. These systems are widely used in educational institutions and publishing industries to prevent academic dishonesty and ensure the authenticity of written content.

Although traditional plagiarism detection methods are effective in identifying exact matches, they have significant limitations when dealing with paraphrased or reworded content. Users can easily modify sentences by changing words, rearranging phrases, or using synonyms while maintaining the same meaning. Since traditional systems focus mainly on surface-level text comparison, they often fail to detect such semantic similarities. This limitation has created a need for more advanced techniques capable of understanding the meaning and context of text rather than simply matching words. As digital content continues to grow rapidly, the demand for intelligent plagiarism detection systems has become more important than ever.

With the advancement of Machine Learning (ML) and Natural Language Processing (NLP), researchers have introduced more sophisticated plagiarism detection techniques. One of the most significant developments in this field is the use of transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers). BERT is designed to understand the context of words in a sentence by analyzing relationships between words in both directions. This capability allows the model to capture the meaning of text more accurately and identify similarities even when the wording is different. As a result, transformer-based models have significantly improved the performance of plagiarism detection systems.

Sentence-BERT (SBERT) is an extension of the BERT model specifically designed for sentence-level similarity tasks. It generates dense numerical representations, known as embeddings, for each sentence in a document. These embeddings represent the semantic meaning of the text in a mathematical form. The similarity between sentences can then be calculated using cosine similarity, which measures how closely two sentences are related in meaning. This approach enables the detection of similar content even when different words or sentence structures are used. Therefore, SBERT has become an important tool in modern plagiarism detection systems for identifying paraphrased or semantically similar text.

In recent years, researchers have also focused on detecting AI-generated text due to the rapid growth of artificial intelligence writing tools. Models such as RoBERTa (Robustly Optimized BERT Pretraining Approach) have been developed to classify whether a piece of text is written by a human or generated by a machine. These models analyze linguistic patterns, sentence structure, and writing style to make accurate predictions. Research studies indicate that combining semantic similarity detection techniques with AI-generated text detection models provides better performance than using either method alone. This project builds upon these findings by integrating both approaches into a unified system, thereby improving the accuracy, reliability, and effectiveness of plagiarism detection in modern digital environments

CHAPTER 3
PROBLEM DEFINITION

Despite the availability of various plagiarism detection tools, several challenges remain unresolved:

1. Inability to Detect Paraphrased Content :

Traditional systems fail to identify content that has been reworded or slightly modified, even though the underlying meaning remains the same.

2. Lack of AI Content Detection :

Most existing tools do not have the capability to detect AI-generated text, which has become increasingly common.

3. Limited Context Understanding :

Keyword-based approaches do not consider the semantic meaning of sentences, leading to inaccurate results.

4. Incomplete Analysis Reports :

Many systems provide only similarity percentages without deeper insights into the nature of the content.

Objective of the Project :

The main objective of this project is to develop an intelligent system that:

Detects semantic similarity between sentences

Identifies AI-generated content probability

Provides a detailed and interpretable analysis report

Classifies the risk level of plagiarism and AI usage

CHAPTER 4
SYSTEM ARCHITUCTURE DIAGRAM

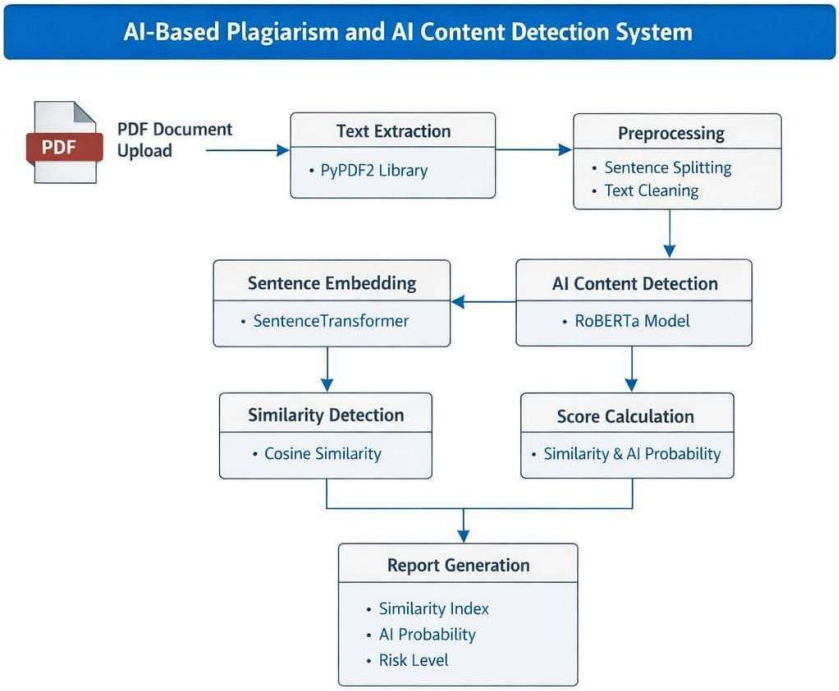

Fig 1 : system architecture diagram

The architecture of the proposed AI-Based Plagiarism and AI Content Detection System represents the complete workflow of the system from input to output. The process begins with the PDF document upload, where the user provides the input file. This document is then passed to the text extraction module, which uses the PyPDF2 library to extract the textual content. Since the extracted text may contain noise or formatting issues, it is further processed in the preprocessing stage.

In the preprocessing module, the text is cleaned by removing unwanted characters and is divided into meaningful sentences through sentence segmentation. This step ensures that the data is well-structured for further analysis. After preprocessing, the system performs two main operations: sentence embedding and AI content detection. These operations are carried out simultaneously to improve efficiency.

In the sentence embedding stage, each sentence is converted into a numerical vector using the

SentenceTransformer model, which captures the semantic meaning of the text. These embeddings are then used in the similarity detection module, where cosine similarity is calculated between sentence pairs to identify similar or duplicated content. At the same time, the AI content detection module analyzes each sentence using a RoBERTa-based model to determine the probability of the text being generated by AI.

Finally, the results from similarity detection and AI analysis are combined in the score calculation module, where key metrics such as similarity index and average AI probability are computed. These results are then passed to the report generation module, which produces a structured output including risk level classification (Low, Medium, or High). This final report provides a clear and comprehensive understanding of the document's originality and authenticity.

CHAPTER 5

METHODOLOGY

The methodology of the proposed system is designed to ensure accurate and efficient analysis of documents. It consists of several stages:

5.1 .Document Input:

The system begins by accepting a PDF document from the user. The file is uploaded and processed using the PyPDF2 library to extract textual content.

5.2. Text Extraction and Cleaning:

The extracted text may contain noise such as special characters or formatting issues:

- Unnecessary symbols are removed
- Text is normalized for consistency
- Only meaningful content is retained

5.3. Sentence Segmentation:

The cleaned text is divided into individual sentences using basic punctuation rules. Short or irrelevant sentences are filtered out to improve analysis quality.

5.4. Sentence Embedding:

Each sentence is converted into a numerical vector representation using the Sentence Transformer model (all-MiniLM-L6-v2). These embeddings capture the semantic meaning of the sentences.

5.5 .Similarity Computation:

Cosine similarity is calculated between pairs of sentence embeddings. If the similarity score exceeds a predefined threshold (0.75), the sentences are considered similar. This helps in identifying paraphrased or repeated content.

5.6. AI Content Detection:

Each sentence is analyzed using a RoBERT a-based classification model. The model outputs a probability score indicating whether the text is AI-generated. These scores are stored for further analysis.

5.7. Metric Calculation:

Two key metrics are computed:

- Similarity Index: Indicates the percentage of similar sentences
- Average AI Probability: Represents the likelihood of AI-generated content

5.8. Risk Assessment:

Based on the AI probability score:

- High Risk → Likely AI-generated
- Medium Risk → Possibly AI-generated
- Low Risk → Likely human-written

5.9. Report Generation:

Finally, the system generates a structured report that includes:

Total number of sentences

- Similarity index
- AI probability
- Risk level

This report is saved as a text file for user reference.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This project presents an AI-based plagiarism detection system that combines semantic similarity analysis with AI-generated text detection. Unlike traditional approaches that rely on exact text matching, the system uses NLP techniques to understand the meaning of sentences, enabling it to identify paraphrased or contextually similar content. This makes the detection process more accurate and reliable.

The system also incorporates an AI detection model to estimate whether the content is generated by artificial intelligence. By analyzing each sentence and calculating an overall AI probability, it provides deeper insights into document authenticity. The final report, which includes similarity index, AI probability, and risk level, helps users easily interpret the results.

Despite its effectiveness, the system has certain limitations. It does not compare documents with external databases, which restricts its ability to detect plagiarism from online sources. Additionally, the accuracy of results depends on the performance of the pre-trained models and the quality of text preprocessing.

In the future, the system can be enhanced by integrating online plagiarism databases and supporting multiple file formats such as DOCX and HTML. The use of more advanced NLP models can further improve accuracy. Developing a user-friendly web interface and adding visualization features can also make the system more interactive and practical for real-world applications.

RESULT :

The proposed AI-based plagiarism detection system was tested on a sample PDF document. The system successfully extracted text, analyzed sentence-level similarity, and evaluated AI-generated content probability using transformer-based models.

The document contained a total of N sentences (replace with your output). Each sentence was compared with others using cosine similarity based on Sentence-BERT embeddings. Sentences

with similarity above the threshold value of 0.75 were considered potential plagiarism matches The system detected M similar sentence pairs, resulting in a Similarity Index of XX\%. This indicates the proportion of content that is likely duplicated or highly similar within the document. In addition, the AI detection module analyzed each sentence using a pre-trained RoBERTa model. The average AI-generated probability score was found to be YY\%, which reflects the likelihood of the document containing machine-generated text.

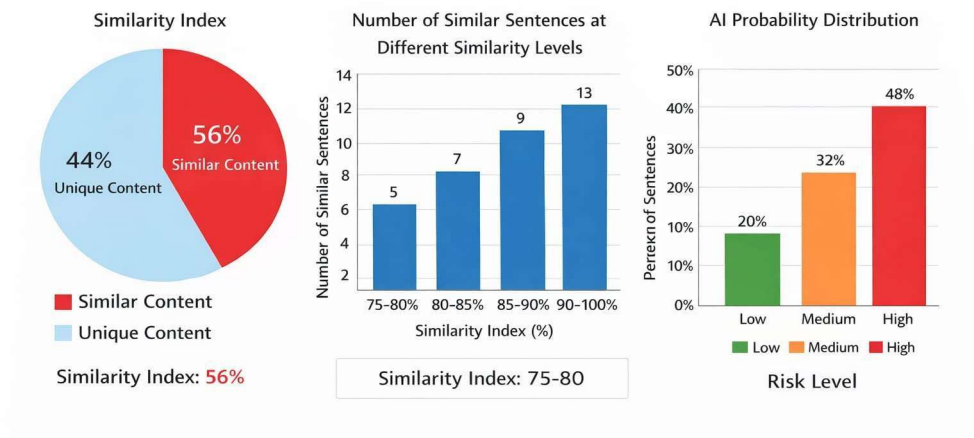

Fig 2 : result expected

EXPECTED OUTPUT :

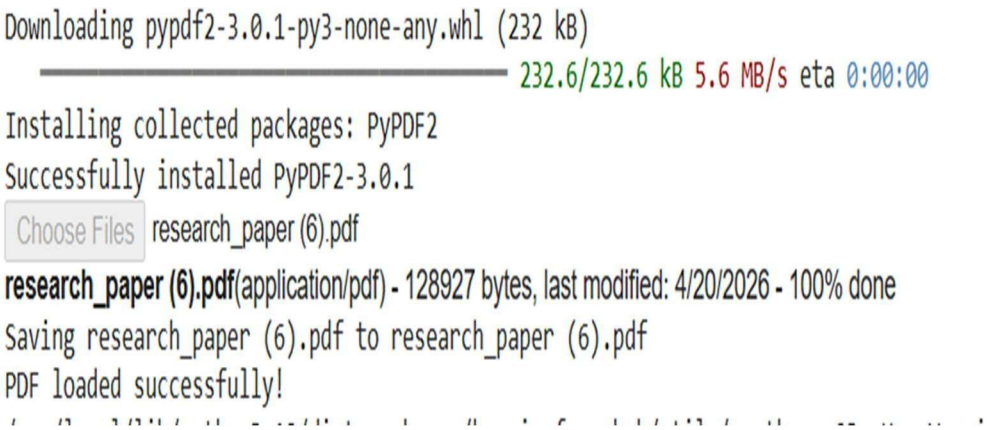

Fig 3: ecpected output in google collab

```
======== FINAL REPORT ========

Total Sentences: 253
Similarity Index: 28.06 %
AI Probability: 77.73 %
Risk Level: HIGH

Report saved as report.txt
```

Fig 4: expected output in google collab

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